

Derivation of Symmetrized Maxwell's Equations

Incorporating Magnetic Monopoles via Dual 4-Potentials

1. Introduction

In standard classical electrodynamics, the absence of magnetic monopoles is embedded in the theory by deriving the electromagnetic fields from a single 4-vector potential, $A^\mu = (\varphi, \mathbf{A})$. Because the magnetic field is defined as the curl of a vector potential ($\mathbf{B} = \nabla \times \mathbf{A}$), its divergence is identically zero ($\nabla \cdot \mathbf{B} = 0$).

To mathematically symmetrize Maxwell's equations and allow for magnetic monopoles, we must introduce a **second** 4-vector potential specifically for the magnetic sector.

2. The Dual Potentials and Field Definitions

Let us define the electric and magnetic 4-vector potentials, along with their respective charge and current densities:

- **Electric Potential:** $A_e^\mu = (\varphi_e, \mathbf{A}_e)$, driven by electric charge ρ_e and current $\mathbf{J}_e$.
- **Magnetic Potential:** $A_m^\mu = (\varphi_m, \mathbf{A}_m)$, driven by magnetic charge ρ_m and current $\mathbf{J}_m$.

Both potentials must independently satisfy the Lorenz gauge condition:

$$\nabla \cdot \mathbf{A}_e + \frac{1}{c} \frac{\partial \varphi_e}{\partial t} = 0, \quad \nabla \cdot \mathbf{A}_m + \frac{1}{c} \frac{\partial \varphi_m}{\partial t} = 0 \quad (1)$$

Furthermore, we postulate that both potentials satisfy the inhomogeneous wave equations (in CGS units):

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \varphi_{e,m} = -4\pi \rho_{e,m} \quad (2)$$

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \mathbf{A}_{e,m} = -\frac{4\pi}{c} \mathbf{J}_{e,m} \quad (3)$$

To symmetrize the theory, the electric and magnetic fields are redefined such that they receive complementary contributions from both potentials:

$$\mathbf{E} = -\nabla \varphi_e - \frac{1}{c} \frac{\partial \mathbf{A}_e}{\partial t} - \nabla \times \mathbf{A}_m \quad (4)$$

$$\mathbf{B} = \nabla \times \mathbf{A}_e - \nabla \varphi_m - \frac{1}{c} \frac{\partial \mathbf{A}_m}{\partial t} \quad (5)$$

3. Deriving the Gauss's Laws

3.1. Gauss's Law for Magnetism

To find the divergence of the magnetic field, we take the divergence of Equation 5:

$$\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}_e) - \nabla^2 \varphi_m - \frac{1}{c} \frac{\partial}{\partial t} (\nabla \cdot \mathbf{A}_m) \quad (6)$$

The divergence of a curl is zero ($\nabla \cdot (\nabla \times \mathbf{A}_e) = 0$). We can substitute the divergence of $\mathbf{A}_m$ using the Lorenz gauge condition from Equation 1 ($\nabla \cdot \mathbf{A}_m = -\frac{1}{c} \frac{\partial \varphi_m}{\partial t}$):

$$\begin{aligned}\nabla \cdot \mathbf{B} &= -\nabla^2 \varphi_m - \frac{1}{c} \frac{\partial}{\partial t} \left(-\frac{1}{c} \frac{\partial \varphi_m}{\partial t} \right) \\ &= -\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \varphi_m\end{aligned}\tag{7}$$

Using the scalar wave equation Equation 2 for the magnetic potential, we recover the symmetrized Gauss's Law for magnetism:

$$\nabla \cdot \mathbf{B} = 4\pi\rho_m\tag{8}$$

3.2. Gauss's Law (Electric)

By an identical mathematical procedure, taking the divergence of Equation Equation 4 yields:

$$\nabla \cdot \mathbf{E} = -\nabla^2 \varphi_e - \frac{1}{c} \frac{\partial}{\partial t} (\nabla \cdot \mathbf{A}_e) - \nabla \cdot (\nabla \times \mathbf{A}_m)\tag{9}$$

Applying the Lorenz gauge and the scalar wave equation for the electric potential gives:

$$\nabla \cdot \mathbf{E} = 4\pi\rho_e\tag{10}$$

4. Deriving the Curl Equations

4.1. Symmetrized Faraday's Law

We take the curl of the electric field definition Equation 4:

$$\nabla \times \mathbf{E} = -\nabla \times (\nabla \varphi_e) - \frac{1}{c} \frac{\partial}{\partial t} (\nabla \times \mathbf{A}_e) - \nabla \times (\nabla \times \mathbf{A}_m)\tag{11}$$

The curl of a gradient is zero ($\nabla \times \nabla \varphi_e = 0$). Next, we isolate $\nabla \times \mathbf{A}_e$ from the definition of the magnetic field Equation 5:

$$\nabla \times \mathbf{A}_e = \mathbf{B} + \nabla \varphi_m + \frac{1}{c} \frac{\partial \mathbf{A}_m}{\partial t}\tag{12}$$

Substituting this into our equation for $\nabla \times \mathbf{E}$:

$$\begin{aligned}\nabla \times \mathbf{E} &= -\frac{1}{c} \frac{\partial}{\partial t} \left(\mathbf{B} + \nabla \varphi_m + \frac{1}{c} \frac{\partial \mathbf{A}_m}{\partial t} \right) - \nabla \times (\nabla \times \mathbf{A}_m) \\ &= -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} - \frac{1}{c} \nabla \frac{\partial \varphi_m}{\partial t} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}_m}{\partial t^2} - \nabla \times (\nabla \times \mathbf{A}_m)\end{aligned}\tag{13}$$

We use the vector identity $\nabla \times (\nabla \times \mathbf{A}_m) = \nabla(\nabla \cdot \mathbf{A}_m) - \nabla^2 \mathbf{A}_m$:

$$\nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = -\frac{1}{c} \nabla \frac{\partial \varphi_m}{\partial t} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}_m}{\partial t^2} - \nabla(\nabla \cdot \mathbf{A}_m) + \nabla^2 \mathbf{A}_m\tag{14}$$

Grouping the gradient terms and the $\mathbf{A}_m$ terms together:

$$\nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = \left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \mathbf{A}_m - \nabla \left(\nabla \cdot \mathbf{A}_m + \frac{1}{c} \frac{\partial \varphi_m}{\partial t} \right)\tag{15}$$

By the Lorenz gauge Equation 1, the gradient term vanishes entirely. We are left with the wave operator acting on $\mathbf{A}_m$. Substituting the vector wave equation Equation 3 gives the symmetrized Faraday's Law:

$$\nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = -\frac{4\pi}{c} \mathbf{J}_m \quad (16)$$

4.2. Symmetrized Ampere-Maxwell Law

Finally, we take the curl of the magnetic field definition Equation 5:

$$\nabla \times \mathbf{B} = \nabla \times (\nabla \times \mathbf{A}_e) - \nabla \times (\nabla \varphi_m) - \frac{1}{c} \frac{\partial}{\partial t} (\nabla \times \mathbf{A}_m) \quad (17)$$

The second term vanishes. We isolate $\nabla \times \mathbf{A}_m$ from the definition of the electric field Equation 4:

$$-\nabla \times \mathbf{A}_m = \mathbf{E} + \nabla \varphi_e + \frac{1}{c} \frac{\partial \mathbf{A}_e}{\partial t} \quad (18)$$

Substituting this into the $\nabla \times \mathbf{B}$ equation:

$$\begin{aligned} \nabla \times \mathbf{B} &= \nabla \times (\nabla \times \mathbf{A}_e) + \frac{1}{c} \frac{\partial}{\partial t} \left(\mathbf{E} + \nabla \varphi_e + \frac{1}{c} \frac{\partial \mathbf{A}_e}{\partial t} \right) \\ &= \nabla (\nabla \cdot \mathbf{A}_e) - \nabla^2 \mathbf{A}_e + \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} + \frac{1}{c} \nabla \frac{\partial \varphi_e}{\partial t} + \frac{1}{c^2} \frac{\partial^2 \mathbf{A}_e}{\partial t^2} \end{aligned} \quad (19)$$

Rearranging and moving the time derivative of $\mathbf{E}$ to the left side:

$$\nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = - \left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \mathbf{A}_e + \nabla \left(\nabla \cdot \mathbf{A}_e + \frac{1}{c} \frac{\partial \varphi_e}{\partial t} \right) \quad (20)$$

Once again, the Lorenz gauge Equation 1 forces the gradient term to zero. Applying the vector wave equation Equation 3 for $\mathbf{A}_e$, we obtain the standard Ampere-Maxwell law:

$$\nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = \frac{4\pi}{c} \mathbf{J}_e \quad (21)$$

5. Summary

By introducing a dual set of 4-potentials, the electric and magnetic fields are placed on completely equal footing. Equations Equation 8, Equation 10, Equation 16, and Equation 21 form the beautifully symmetric Maxwell's equations.